G. Leigh Harwell

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Academic Website
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Education

University of Oklahoma August 2023 – May 2027
Bachelor of Science in Astrophysics | Minor in Mathematics | Data Analytics Certificate *Norman, OK, USA*

Research Experience

Multi-Messenger Observations of Compact Objects August, 2026 – Present
University of Oklahoma, Department of Physics and Astronomy *Norman, OK, USA*

- Characterizing double white dwarf systems through multi messenger observations.
- Analyze multi messenger datasets to identify compact binary candidates for follow-up observational studies.

Machine Learning and Stellar Parameter Estimations October, 2024 – October 2026
University of Oklahoma, Department of Physics and Astronomy *Norman, OK, USA*

- Transitioned project into an intensive NSF REU at the University of Oklahoma during Summer 2026, focusing on advanced stellar parameter estimation.
- Apply machine learning techniques to asteroseismology and galactic archaeology datasets to improve stellar classification models.
- Implementing "TheCannon" data driven model to estimate stellar parameters in Astroseismolog with elemental abundances from spectroscopic data.

Other Experience

Physics for Life Science Majors Tutoring Coordinator & PLA August, 2025 – Present
University of Oklahoma, Student Learning Center, Action Tutoring *Norman, OK, USA*

- Coordinating the tutoring for the life science majors, and collaborating with department faculty to provide extra problem sets and conceptual worksheets for exam review.
- Provide individual and small-group tutoring sessions in introductory mechanics, optics, thermodynamics, and electromagnetism.

Research Presentations

Characterizing Stellar Labels with Machine Learning 2026
Department Research Symposium, University of Oklahoma *Norman, OK, USA*

- **Harwell, G. L.**, Beaty, M., Hayden
- Poster presentation detailing the findings from the UReCA Fellowship program over my research on machine learning and stellar parameter estimations.

Multi-Messenger Observations of Compact Binary Systems 2026
Capstone Research Presentation, University of Oklahoma *Norman, OK, USA*

- **Harwell, G. L.**, Barrientos, M., Kilic, M.
- Five minute oral presentation over the back ground for my capstone research on multi-messenger observations.

Awards & Honors

URECA Fellowship 2026

The University of Oklahoma *Norman, OK, USA*

- Funded collaborative research developing machine learning tools for stellar label estimation; presented findings at department symposium.

Physics and Astronomy Meritorious Scholarship 2024 – Present

University of Oklahoma, Homer L. Dodge Department *Norman, OK, USA*

- Awarded to outstanding physics and astronomy students that maintain high academic standing.

President's Honor Roll 2024 – Present

University of Oklahoma *Norman, OK, USA*

- Awarded to distinguished students that maintain a 4.0 GPA throughout the semester.

Specialized Skills

Programming Languages: Python (intermediate)(Scientific Libraries: NumPy, SciPy, Pandas, Matplotlib, AstroPy, and scikit-learn), LaTeX (beginner)
Data Analysis: Multi-messenger data analysis, stellar parameter modeling, and machine learning workflow
Computer Literacy: Utilize various tools in the operation of computers and various other technology
Adaptability: Respond flexibly to evolving factors, environments, and conditions by using new skills

Extracurricular

Community and Belonging Committee| August 2026 – Present
Society of Physics Students| August 2023 – Present
Women in Physics| August 2023 – Present